

Mathematical Foundations of Machine Learning

Assignment 1 Solutions

Question 1 (20 points)

We have the following system:

$$\mathbf{Ax} = 4\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^3, \quad \sum_{i=1}^3 x_i = 1,$$

where

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 1 \\ 2 & 2 & 2 \\ 1 & 0 & 3 \end{bmatrix}.$$

Describe all solutions $\mathbf{x} = [x_1 \ x_2 \ x_3]^\top$ that satisfy $\mathbf{Ax} = 4\mathbf{x}$ with the constraint $x_1 + x_2 + x_3 = 1$.

Solution:

1. **Set up the equation:** $(A - 4I)x = 0$

$$\begin{bmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

2. **Reduce the matrix:** Performing row operations leads to the system:

$$\begin{aligned} x_1 - x_3 &= 0 \implies x_1 = x_3 \\ x_2 - 2x_3 &= 0 \implies x_2 = 2x_3 \end{aligned}$$

3. **Apply the constraint:** Substitute $x_1 = x_3$ and $x_2 = 2x_3$ into $x_1 + x_2 + x_3 = 1$:

$$x_3 + 2x_3 + x_3 = 1 \implies 4x_3 = 1 \implies x_3 = \frac{1}{4}$$

4. **Final Answer:**

$$x = \begin{bmatrix} 1/4 \\ 1/2 \\ 1/4 \end{bmatrix}$$

Question 2 (20 points)

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$.

1. (8 points) Calculate B^{-1} by augmenting it with the identity matrix I_2 and transforming it into $[I_2|B^{-1}]$.
2. (6 points) Calculate $(AB)^{-1}$. Does $(AB)^{-1} = A^{-1}B^{-1}$?
3. (6 points) Determine if $(A + I_2)^{-1} = A^{-1} + I_2^{-1}$ (I_2 is the identity matrix).

Solution:

$$1. [B|I] = \left[\begin{array}{cc|cc} 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1/2} \left[\begin{array}{cc|cc} 1 & 0 & 0.5 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 0 & 0.5 & 0 \\ 0 & 1 & -0.5 & 1 \end{array} \right]$$

$$B^{-1} = \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix}$$

$$2. AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

$$(AB)^{-1} = \frac{1}{3-1} \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix}$$

However, $A^{-1}B^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -0.5 & 1 \end{bmatrix}$. They are not equal because $(AB)^{-1} = B^{-1}A^{-1}$.

$$3. A + I_2 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \implies (A + I_2)^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.25 \\ 0 & 0.5 \end{bmatrix}.$$

$$A^{-1} + I_2^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$$

They are not equal. Generally, $(A + B)^{-1} \neq A^{-1} + B^{-1}$.

Question 3 (20 points)

Consider the following system of linear equations in variables x, y, z which depends on a real parameter k :

$$\begin{aligned} x + 2y + kz &= 1 \\ 2x + ky + 8z &= 3 \\ x + (k-2)y + (8-k)z &= 2 \end{aligned}$$

- (6 points) Write the augmented matrix $[A|b]$ and use elementary row operations to reduce it to row-echelon form.
- (6 points) Determine all values of k for which the system has:
 - (i) A unique solution.
 - (ii) Infinitely many solutions.
 - (iii) No solution.
- (8 points) For $k = 5$ that results in infinitely many solutions, express the solution set in parametric form.

Solution:

- The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 2 & k & 1 \\ 2 & k & 8 & 3 \\ 1 & k-2 & 8-k & 2 \end{array} \right] \xrightarrow[\substack{R_2-2R_1 \\ R_3-R_1}]{} \left[\begin{array}{ccc|c} 1 & 2 & k & 1 \\ 0 & k-4 & 8-2k & 1 \\ 0 & k-4 & 8-2k & 1 \end{array} \right] \xrightarrow{R_3-R_2} \left[\begin{array}{ccc|c} 1 & 2 & k & 1 \\ 0 & k-4 & 8-2k & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

- From the row-echelon form:

- Unique solution:** Impossible. The third row is $0 = 0$, so the rank of the augmented matrix is at most 2, while we have 3 variables.
- Infinitely many solutions:** Occurs if the system is consistent. Since the last row is $0 = 0$, the system is consistent as long as the second row does not lead to a contradiction.
- No solution:** Occurs if $k = 4$.

- Let $k = 5$. The row-echelon form is $[1 \ 2 \ 5 \mid 1]$ and $[0 \ 1 \ -2 \mid 1]$. From R_2 : $y - 2t = 1 \implies y = 1 + 2t$. From R_1 : $x + 2(1 + 2t) + 5t = 1 \implies x = -1 - 9t$.
Solution set: $\{(-1 - 9t, 1 + 2t, t) \mid t \in \mathbb{R}\}$.

Question 4 (15 points)

Consider the set $\mathcal{G} = \{x \in \mathbb{R} \mid x > 0, x \neq 1\}$ with the operation $*$ defined by:

$$a * b := e^{(\ln a \cdot \ln b)}, \quad \forall a, b \in \mathcal{G}$$

- (5 points) Solve for x in the expression:

$$x * e * e^2 = e^8$$

(where e is Euler's number).

- (10 points) Show that $(\mathcal{G}, *)$ is an Abelian group.

Solution:

1. We interpret the expression from left to right:

$$(x * e) * e^2 = e^8.$$

First,

$$x * e = e^{(\ln x)(\ln e)} = e^{\ln x} = x.$$

Hence

$$x * e^2 = e^8.$$

By the definition of $*$,

$$e^{(\ln x)(\ln e^2)} = e^8 \implies e^{2 \ln x} = e^8 \implies 2 \ln x = 8.$$

Therefore

$$\ln x = 4 \implies x = e^4.$$

2. We verify the group axioms for $(\mathcal{G}, *)$.

- **Closure:** If $a, b \in \mathcal{G}$, then $a > 0$, $b > 0$, and $a \neq 1$, $b \neq 1$. Hence $\ln a \neq 0$ and $\ln b \neq 0$, so

$$\ln a \cdot \ln b \neq 0.$$

Therefore

$$a * b = e^{(\ln a)(\ln b)} > 0 \quad \text{and} \quad a * b \neq 1.$$

Thus $a * b \in \mathcal{G}$.

- **Associativity:** For any $a, b, c \in \mathcal{G}$,

$$(a * b) * c = e^{\ln(a*b) \ln c} = e^{((\ln a)(\ln b)) \ln c},$$

while

$$a * (b * c) = e^{\ln a \ln(b*c)} = e^{\ln a ((\ln b)(\ln c))}.$$

Since real multiplication is associative,

$$((\ln a)(\ln b)) \ln c = \ln a ((\ln b)(\ln c)),$$

so

$$(a * b) * c = a * (b * c).$$

- **Neutral element:** We seek $e_{\mathcal{G}} \in \mathcal{G}$ such that for every $a \in \mathcal{G}$,

$$a * e_{\mathcal{G}} = a.$$

This means

$$e^{(\ln a)(\ln e_{\mathcal{G}})} = a = e^{\ln a}.$$

Since $\ln a \neq 0$, we get

$$\ln e_{\mathcal{G}} = 1,$$

hence

$$e_{\mathcal{G}} = e.$$

- **Inverse element:** Given $a \in \mathcal{G}$, we seek $b \in \mathcal{G}$ such that

$$a * b = e.$$

Then

$$e^{(\ln a)(\ln b)} = e \implies (\ln a)(\ln b) = 1 \implies \ln b = \frac{1}{\ln a}.$$

Therefore

$$b = e^{1/\ln a}.$$

This is well-defined because $\ln a \neq 0$. Also $b > 0$, and $b \neq 1$ because $1/\ln a \neq 0$. Thus $b \in \mathcal{G}$.

- **Commutativity:** For any $a, b \in \mathcal{G}$,

$$a * b = e^{(\ln a)(\ln b)} = e^{(\ln b)(\ln a)} = b * a.$$

Therefore $(\mathcal{G}, *)$ is an Abelian group.

Question 5 (15 points)

Let $\mathcal{V} = \mathbb{R}^2$. Define the operations addition $(\oplus)$ and scalar multiplication $(\odot)$ for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\alpha \in \mathbb{R}$ as:

$$\mathbf{u} \oplus \mathbf{v} = (u_1 + v_1 + 1, u_2 + v_2 + 1)$$

$$\alpha \odot \mathbf{u} = (\alpha u_1 + \alpha - 1, \alpha u_2 + \alpha - 1)$$

Given that $(\mathcal{V}, \oplus)$ forms an Abelian group:

1. (5 points) Find the neutral element and the inverse element of $(\mathcal{V}, \oplus)$.
2. (10 points) Determine whether $(\mathcal{V}, \oplus, \odot)$ is a vector space. Please justify your answer.

Solution:

Part 1

- **Neutral element:** $\mathbf{u} \oplus \mathbf{e} = \mathbf{u}$: $u_i + z_i + 1 = u_i \implies z_i = -1$. Thus the neutral element is $[-1, -1]$.
- **Inverse element:** $\mathbf{u} \oplus \mathbf{w} = [-1, -1]$: $u_i + w_i + 1 = -1 \implies w_i = -u_i - 2$. Thus the inverse element is $[-u_1 - 2, -u_2 - 2]$.

Part 2

Given $(\mathcal{V}, \oplus)$ is an Abelian group, we verify the properties for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{R}$:

- **Distributivity (Scalar):** $\alpha \odot (\mathbf{u} \oplus \mathbf{v}) = [\alpha(u_1 + v_1 + 1) + \alpha - 1, \alpha(u_2 + v_2 + 1) + \alpha - 1] = [\alpha u_1 + \alpha v_1 + 2\alpha - 1, \alpha u_2 + \alpha v_2 + 2\alpha - 1]$.
 $(\alpha \odot \mathbf{u}) \oplus (\alpha \odot \mathbf{v}) = [(\alpha u_1 + \alpha - 1) + (\alpha v_1 + \alpha - 1) + 1, (\alpha u_2 + \alpha - 1) + (\alpha v_2 + \alpha - 1) + 1] = [\alpha u_1 + \alpha v_1 + 2\alpha - 1, \alpha u_2 + \alpha v_2 + 2\alpha - 1]$.
- **Distributivity (Vector):** $(\alpha + \beta) \odot \mathbf{u} = [(\alpha + \beta)u_1 + \alpha + \beta - 1, (\alpha + \beta)u_2 + \alpha + \beta - 1]$.
 $(\alpha \odot \mathbf{u}) \oplus (\beta \odot \mathbf{u}) = [(\alpha u_1 + \alpha - 1) + (\beta u_1 + \beta - 1) + 1, (\alpha u_2 + \alpha - 1) + (\beta u_2 + \beta - 1) + 1] = [(\alpha + \beta)u_1 + \alpha + \beta - 1, (\alpha + \beta)u_2 + \alpha + \beta - 1]$.
- **Associativity:** $\alpha \odot (\beta \odot \mathbf{u}) = \alpha \odot [\beta u_1 + \beta - 1, \beta u_2 + \beta - 1] = [\alpha(\beta u_1 + \beta - 1) + \alpha - 1, \alpha(\beta u_2 + \beta - 1) + \alpha - 1] = [\alpha\beta u_1 + \alpha\beta - 1, \alpha\beta u_2 + \alpha\beta - 1] = (\alpha\beta) \odot \mathbf{u}$.
- **Neutral element with respect to the outer operation:** $1 \odot \mathbf{u} = [1 \cdot u_1 + 1 - 1, 1 \cdot u_2 + 1 - 1] = [u_1, u_2] = \mathbf{u}$.

Since all properties are satisfied, $(\mathcal{V}, \oplus, \odot)$ is a vector space.

Question 6 (10 points)

Determine whether the following subsets are subspaces of $\mathbb{R}^{n \times n}$. Please justify your answers.

1. (5 points) $W_1 = \{A \in \mathbb{R}^{n \times n} \mid A^\top = -A\}$
2. (5 points) $W_2 = \{A \in \mathbb{R}^{n \times n} \mid \sum_{i=1}^n A_{i,i} = 1\}$

Solution:

1. $W_1 = \{A \in \mathbb{R}^{n \times n} \mid A^\top = -A\}$ is a subspace of $\mathbb{R}^{n \times n}$.

Indeed:

- The zero matrix belongs to W_1 since $\mathbf{0}^\top = -\mathbf{0}$.
- If $A, B \in W_1$, then

$$(A + B)^\top = A^\top + B^\top = -A - B = -(A + B),$$

so $A + B \in W_1$.

- If $\alpha \in \mathbb{R}$ and $A \in W_1$, then

$$(\alpha A)^\top = \alpha A^\top = \alpha(-A) = -(\alpha A),$$

so $\alpha A \in W_1$.

Hence W_1 is a subspace.

2. $W_2 = \{A \in \mathbb{R}^{n \times n} \mid \sum_{i=1}^n A_{i,i} = 1\}$ is not a subspace of $\mathbb{R}^{n \times n}$.

A subspace must contain the zero matrix, but for the zero matrix we have

$$\sum_{i=1}^n 0_{i,i} = 0 \neq 1.$$

So the zero matrix does not belong to W_2 , and therefore W_2 is not a subspace.

Bonus Question (10 points)

Let U and W be subspaces of a vector space V .

- (5 points) Determine if the intersection $U \cap W$ is a subspace of V . Please justify your answer.
- (5 points) Determine if the union $U \cup W$ is a subspace of V . Please justify your answer.

Solution:

1. **Intersection:** $U \cap W$ is a **subspace**. First, since U and W are subspaces, the zero vector $\mathbf{0}$ is in both, so $\mathbf{0} \in U \cap W$ (non-empty).

Second, let $\mathbf{x}, \mathbf{y} \in U \cap W$ and let $\alpha, \beta \in \mathbb{R}$ be scalars. Since $\mathbf{x}, \mathbf{y} \in U$ and U is a subspace, the linear combination $\alpha\mathbf{x} + \beta\mathbf{y} \in U$. Similarly, since $\mathbf{x}, \mathbf{y} \in W$ and W is a subspace, $\alpha\mathbf{x} + \beta\mathbf{y} \in W$. Since the vector is in both U and W , it is in the intersection: $\alpha\mathbf{x} + \beta\mathbf{y} \in U \cap W$.

2. **Union:** $U \cup W$ is **not necessarily a subspace**. We prove this by counterexample. Let $V = \mathbb{R}^2$. Let $U = \{[x, 0] \mid x \in \mathbb{R}\}$, and $W = \{[0, y] \mid y \in \mathbb{R}\}$. Both are valid subspaces.

Consider vector $\mathbf{u} = [1, 0] \in U$ and vector $\mathbf{w} = [0, 1] \in W$. Both vectors are elements of the union $U \cup W$. However, their sum is $\mathbf{u} + \mathbf{w} = [1, 1]$. $[1, 1] \notin U$ because the second component is non-zero, and $[1, 1] \notin W$ because the first component is non-zero. Therefore, $\mathbf{u} + \mathbf{w} \notin U \cup W$. The union is not closed under addition.